Acumatica MCP Developer Guide

Connecting Local IIS Acumatica Instances to Cloudflare Worker MCP Server via Cloudflare Tunnel

DEVELOPER RUNBOOK | MCP 0.53.0 | ACUMATICA XRP | CLOUDFLARE WORKERS

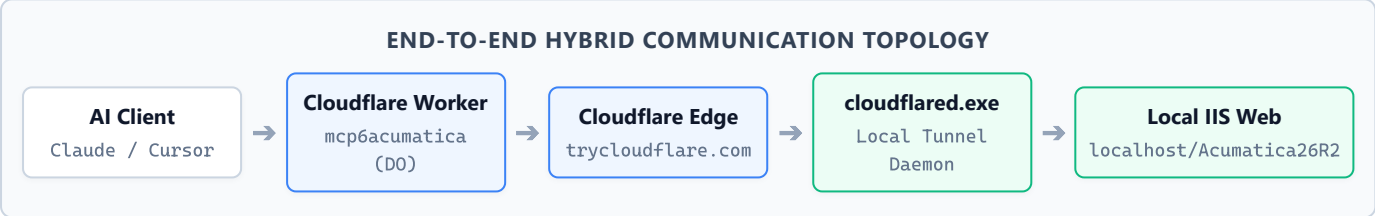
Target Audience: Development Team
Local Environment: Windows IIS / Localhost
Security: OAuth 2.0 / OIDC IdentityServer
Doc Version: 1.0 (Production Ready)

1. Architecture & Why This Tunnel is Needed

When building and testing Acumatica ERP customizations (such as PMS modules, custom DAC extensions, or Generic Inquiries), developers execute code locally against their dedicated IIS development instances (e.g. `http://localhost/Acumatica26R2/`).

The **Model Context Protocol (MCP) Server** runs in the cloud as a Cloudflare Worker (e.g. `mcp6acumatica.inglerohan1808.workers.dev`) and interfaces with AI developer assistants (Claude Desktop, Cursor, Antigravity). Cloudflare Workers operate in a public serverless runtime and cannot reach private loopback addresses (`localhost` or private `192.168.x.x` subnets).

Cloudflare Tunnel (`cloudflared`) solves this by establishing a lightweight, encrypted, outbound-only tunnel from your local machine to Cloudflare's edge network. This exposes your local Acumatica instance over a secure, publicly accessible HTTPS domain without modifying firewall rules, opening router ports, or requiring a static IP.



2. Developer Prerequisites Checklist

- ✓ **Local Acumatica Instance:** Configured in IIS, running and logged in at `http://localhost/<InstanceName>/` (e.g., `http://localhost/Acumatica26R2/`).
- ✓ **Anonymous Authentication Enabled in IIS:** Open IIS Manager ➔ Select your Acumatica Site/Application ➔ Authentication ➔ Ensure **Anonymous Authentication** is `Enabled` (needed for OIDC & REST endpoints).
- ✓ **Node.js v18+ Installed:** Required to run `npx cloudflared`, or install the official `cloudflared.exe` binary.
- ✓ **Acumatica Administrator Access:** Permissions to navigate to screen **Connected Applications (SM303010)** and **Generic Inquiry (SM208000)**.
- ✓ **Cloudflare Account & Worker Access:** Access to the team's Cloudflare Worker dashboard (or personal worker deploy).

1 Launching Cloudflare Tunnel on Your PC

Open a PowerShell terminal on your Windows development workstation and run the Quick Tunnel command:

```
# Start outbound tunnel pointing to your local IIS instance (port 80)
npx cloudflared tunnel --url http://localhost
```

Cloudflare will establish the tunnel and output connection banners:

```
2026-10-23T06:05:12Z INF +--------------------------------------------------------------------------------------------
2026-10-23T06:05:12Z INF |  Your quick Tunnel has been created! Visit it at (it may take some time to be reachable)
2026-10-23T06:05:12Z INF |  https://announcement-auction-classic-magazines.trycloudflare.com
2026-10-23T06:05:12Z INF +--------------------------------------------------------------------------------------------
2026-10-23T06:05:12Z INF Route propagating, please allow up to 20 seconds before accessing this URL...
```

Verifying Tunnel & IdentityServer Health

Open the tunnel URL joined with your Acumatica instance name and IdentityServer discovery endpoint in a browser: `https://<your-tunnel-subdomain>.trycloudflare.com/Acumatica26R2/identity/.well-known/openid-configuration`

Expected Output: A valid JSON payload containing `authorization_endpoint`, `token_endpoint`, and supported scopes. If this responds, your local Acumatica is reachable by Cloudflare!

Developer Rule: Keep the Terminal Running

Quick Tunnels are ephemeral and remain active only as long as this PowerShell process runs. Do not close this terminal while developing.

2 Configuring Acumatica Connected Applications (SM303010)

Acumatica's built-in IdentityServer authenticates users via OAuth 2.0 Authorization Code flow. Configure it as follows:

1. In your local Acumatica instance, search for and open screen **Connected Applications (SM303010)**.
2. Click **+ (Add New Record)** or select an existing MCP application.
3. Fill in the header fields:
   - **Client Name:** `Cloudflare MCP Worker`
   - **OAuth Flow:** `Authorization Code`
4. Under **Secrets**:
   - Click **New Secret**. A popup displays your secret string.
   - **Copy this secret immediately to a secure note** (it cannot be viewed again once closed).
5. Switch to the **REDIRECT URIS** tab:
   - Click **+** and enter your Cloudflare Worker's callback endpoint:
   - `https://mcp6acumatica.inglerohan1808.workers.dev/callback`
6. Switch to the **SCOPES** tab:
   - Ensure scopes `api` and `offline_access` are present and allowed.
7. Click **Save** (Disk icon).
8. Note down your **Client ID**:
   Format: `<GUID>@<TenantName>` (e.g., `6D2DCD41-09F9-6EB8-6147-ED52F132EFE2@Company`).

Field	Value	Purpose
OAuth Flow	`Authorization Code`	Standard interactive user login flow.
Redirect URI	`https://<worker-domain>/callback`	Where IdentityServer sends the authorization code.
Client ID	`<GUID>@<Tenant>`	Identifies the application and login tenant.

3 Updating Cloudflare Worker Variables & Secrets

Now configure your Cloudflare Worker so that it forwards API calls and authentication requests through your local tunnel:

1. Log into the **Cloudflare Dashboard**.
2. Go to **Workers & Pages ➔** Select your MCP Worker (e.g., **`mcp6acumatica`**).
3. Click **Settings ➔ Variables and secrets**.
4. Update the three critical configuration variables:

Variable Name	Type	Value to Set
`ACUMATICA_URL`	Variable	Your tunnel URL + instance virtual directory: `https://announcement-auction-classic-magazines.trycloudflare.com/Acumatica26R2`
`ACUMATICA_CLIENT_ID`	Variable	Your Client ID from SM303010: `6D2DCD41-09F9-6EB8-6147-ED52F132EFE2@Company`
`ACUMATICA_CLIENT_SECRET`	Secret	Click *Edit* or *Add Secret* and paste the secret generated from SM303010.
`ACUMATICA_TENANT`	Variable	Your login tenant name (e.g., `Company`).

> **Deploy & Verify Runtime**
> Click **Save and Deploy**. Cloudflare propagates the new variables to all global edge points within 2-3 seconds.

4 Configuring the Access Gate (MCPAccess Canary GI)

MCP4Acumatica includes an **access gate** to safeguard company ERP data: during login, the server verifies whether the authenticating user has permissions to read a designated canary Generic Inquiry (default name: `MCPAccess`).

1. In Acumatica, navigate to **Generic Inquiry (SM208000)**.
2. Create an inquiry named `MCPAccess` (or your custom name matching `ACUMATICA_CANARY_GI`).
3. Add any simple table (e.g. `Company` or `Branch`) with one field.
4. Check the box: **Expose via OData**.
5. Under **Access Rights by Screen (SM201020)**, ensure your developer user or a marker role (e.g. `MCP Access`) has at least *Granted (Read)* access to this inquiry.

5 Registering MCP in AI Clients & First Login

Add the server endpoint to your AI client's configuration file:

Claude Desktop Configuration (`claude_desktop_config.json`):

```
{
  "mcpServers": {
    "acumatica-local": {
      "url": "https://mcp6acumatica.inglerohan1808.workers.dev/sse"
    }
  }
}
```

Cursor / VS Code MCP Configuration:

Add a new SSE server with URL: `https://mcp6acumatica.inglerohan1808.workers.dev/sse`

First-Time Authentication Walkthrough:

1. Start a conversation in Claude or Cursor and ask: "*List my Acumatica inquiries*" or "*Describe customer schema*".
2. The AI client detects unauthenticated state and launches an OAuth login prompt in your default browser.
3. Notice the address bar points to your secure Cloudflare Tunnel:
   `https://<tunnel-subdomain>.trycloudflare.com/Acumatica26R2/identity/connect/authorize?...`
4. Sign in using your local Acumatica credentials (e.g. `admin`).
5. Click **Authorize / Accept**. You will be redirected back to the Cloudflare Worker callback.
6. Tokens are encrypted and stored in Cloudflare KV, and your AI assistant now has access to your local Acumatica instance!

6 Testing with the Preflight Diagnostic

MCP4Acumatica includes a built-in preflight diagnostic tool to verify all connection layers in one click without guessing:

Navigate in your browser to:
`https://mcp6acumatica.inglerohan1808.workers.dev/docs/admin/preflight`

Log in with your `ADMIN_SECRET`. The diagnostic automatically validates:

- ✓ **Acumatica Reachability:** HTTP ping to the tunnel URL.
- ✓ **OIDC Discovery:** Validates JSON schema from `identity/.well-known/openid-configuration`.
- ✓ **Connected Application Credentials:** Probes token endpoint with Client ID and Secret.
- ✓ **Contract API Endpoint:** Probes `entity/Default/25.200.001` (or configured version).
- ✓ **Canary GI:** Confirms OData read access for `MCPAccess`.

7. Troubleshooting Guide for Developers

Error / Symptom	Probable Root Cause	Resolution Step
Tunnel URL expired or restarted	Quick Tunnels (`npx cloudflared`) generate random domains each run. If the terminal restarts, the URL changes.	Copy the new `.trycloudflare.com` URL and update `ACUMATICA_URL` in Cloudflare Worker settings. Or configure a persistent Named Tunnel.
502 Bad Gateway / Connection Refused	IIS is not listening on `localhost:80` or the Acumatica application pool is stopped.	Open command prompt as admin and run `iisreset`. Verify you can access `http://localhost/Acumatica26R2` directly in your browser.
Redirect URI Mismatch (`invalid_grant`)	The redirect URI in SM303010 doesn't match the Worker URL.	Verify SM303010 contains exactly `https://<worker-domain>/callback` with no trailing slash or whitespace.
401 Unauthorized / Access Gate Denied	The user does not have permission to read the canary GI `MCPAccess`.	In Acumatica SM208000, ensure `MCPAccess` has **Expose via OData** checked and the user has read permissions.
OData Filter Refusal on Inquiries	Acumatica does not support `$filter` on calculated expression columns (columns starting with =).	Filter on stored database fields instead of formula/switch columns, or filter client-side in the AI prompt.

8. Advanced Tip: Setting Up a Permanent Named Tunnel

If you want a static URL that never changes when your machine reboots, set up a free Cloudflare Named Tunnel:

```
# 1. Login to Cloudflare CLI
cloudflared tunnel login

# 2. Create a persistent named tunnel (e.g. dev-acumatica)
cloudflared tunnel create dev-acumatica

# 3. Route your domain (e.g. dev-acumatica.yourdomain.com) to the tunnel
cloudflared tunnel route dns dev-acumatica dev-acumatica.yourdomain.com

# 4. Run the tunnel pointing to localhost
cloudflared tunnel run --url http://localhost dev-acumatica
```

With a named tunnel, your `ACUMATICA_URL` remains constant (e.g. `https://dev-acumatica.yourdomain.com/Acumatica26R2`), eliminating the need to update Cloudflare variables upon terminal restart.